Jet marching methods for numerical geometric acoustics

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Introduction

Introduction: hello!

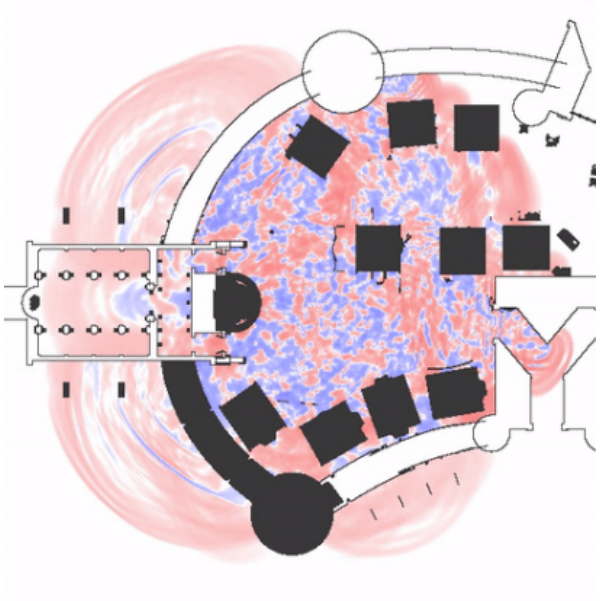
PhD in Computer Science from University of Maryland, advised by Maria Cameron and Ramani Duraiswami.

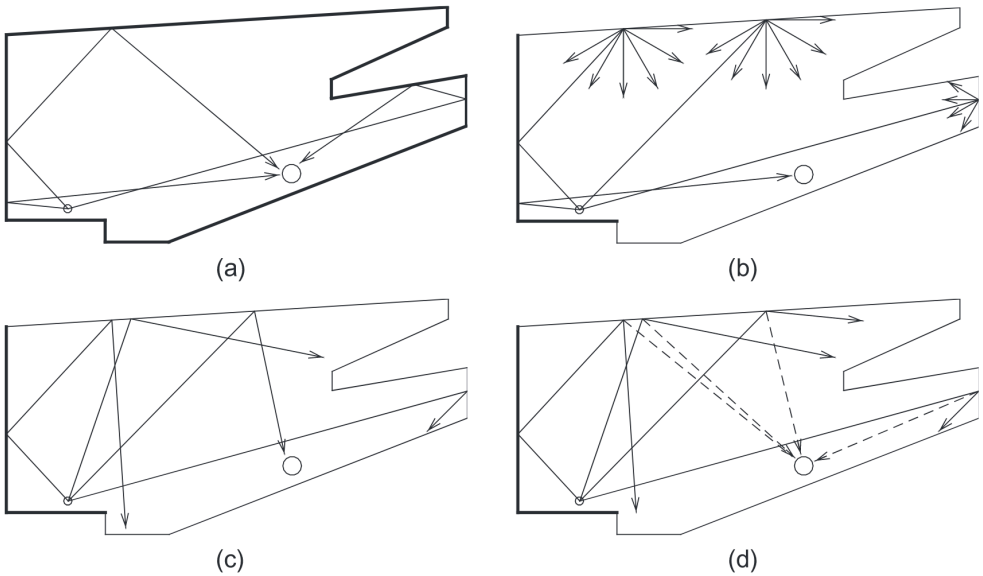
Postdoc at Courant since September 2021 with Leslie Greengard and Mike O'Neil. Working on butterfly algorithms and fast direct solvers.

All material for this talk (software and papers):

`https://sampotter.github.io`

Introduction: computational room acoustics





Numerical acoustics: solve wave equation for bandlimited impulse input. High-frequency (>2kHz) simulations expensive and impractical.

Geometric acoustics: use raytracing to assemble room impulse response. Handling diffraction effects with raytracing is complicated.

Left: `https://www.microsoft.com/en-us/research/project/project-triton`

Right: Kuttruff, 2016.

Introduction: numerical geometric acoustics

Helmholtz equation, where c is the speed of sound and k is the wavenumber:

$$\Delta u_\omega + k^2 u_\omega = 0, \qquad k = \frac{\omega}{c}$$

Geometric optics ansatz ("WKB") for Helmholtz:

$$u_\omega(\boldsymbol{x}) \sim A_\omega(\boldsymbol{x}) e^{i\omega\tau(\boldsymbol{x})}$$

Leads to the eikonal equation:

$$\|\nabla\tau\| = 1/c = s, \qquad s \text{ is the "slowness"}$$

and the amplitude equation:

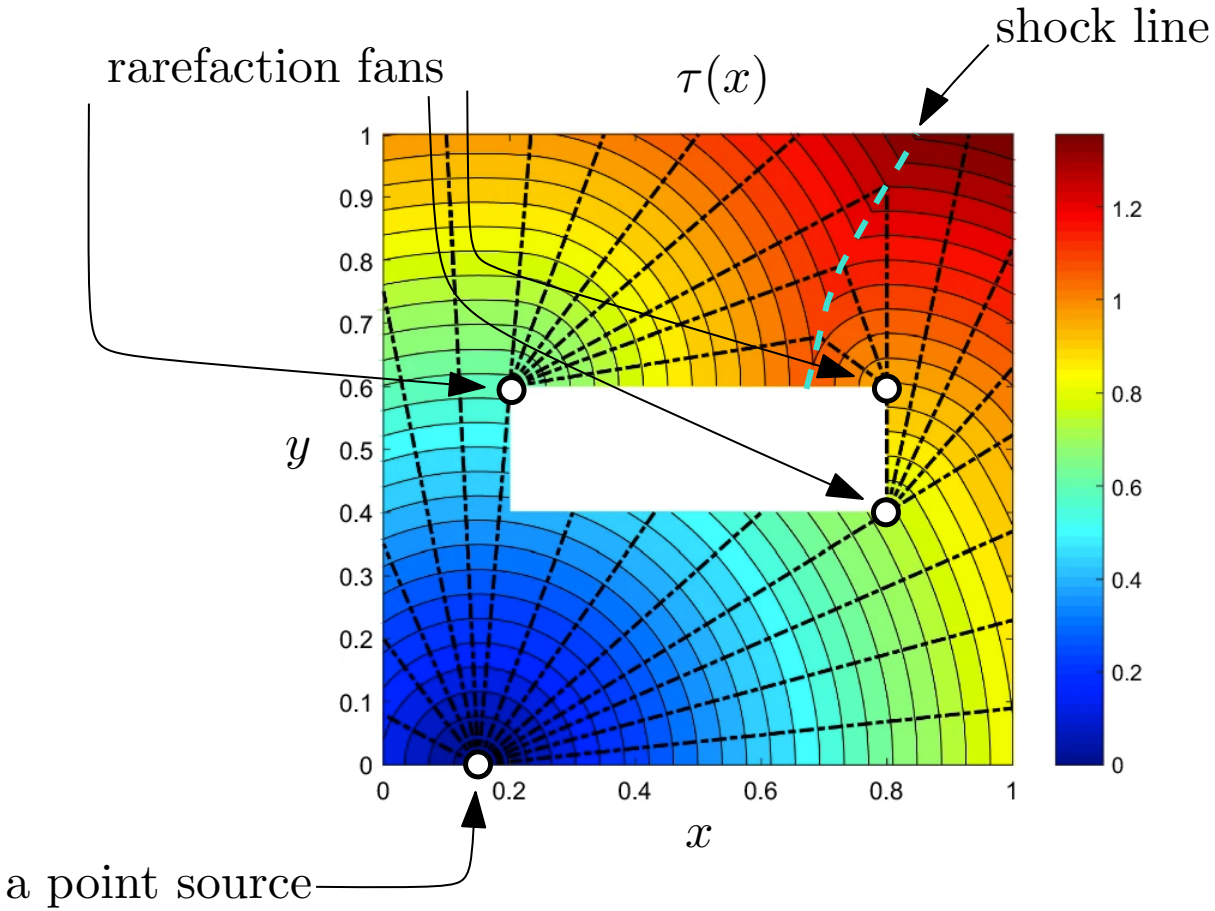
$$\nabla A_\omega \cdot \nabla\tau + 2A_\omega \Delta\tau = 0$$

High-frequency approximation of the Helmholtz equation:

$$|u_\omega - A_\omega e^{i\omega\tau}| = O(\omega^{-1})$$

Introduction: the eikonal equation, $\|\nabla\tau\| = 1/c$

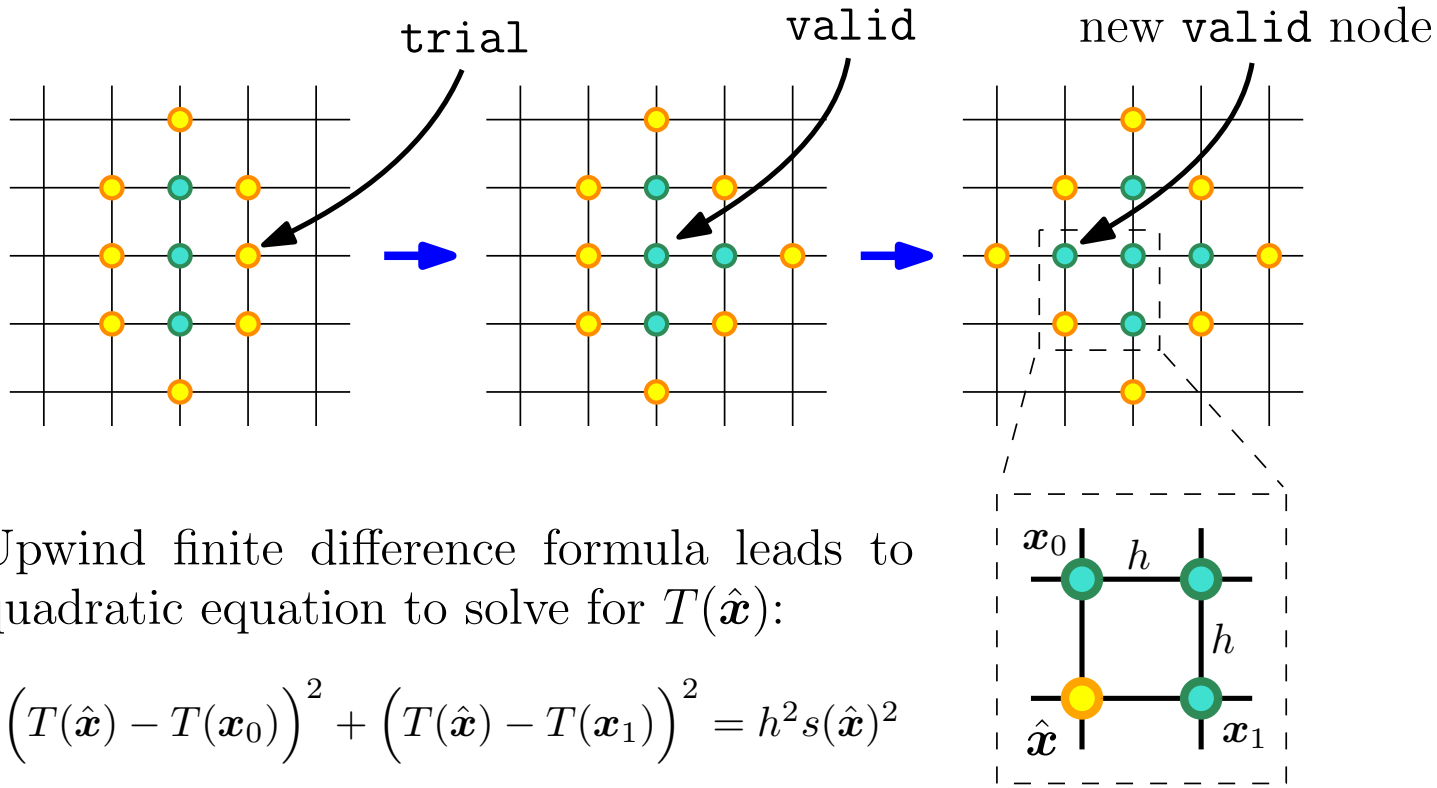
A PDE whose solution encodes only the first arrival time accounting for diffraction. Its characteristics are rays.



Source: Qi & Vladimirsky, 2019.

Introduction: the fast marching method

A single-pass, $O(h)$ accurate, upwind finite difference method for solving the eikonal equation based on Dijkstra's algorithm



Upwind finite difference formula leads to quadratic equation to solve for $T(\hat{\boldsymbol{x}})$:

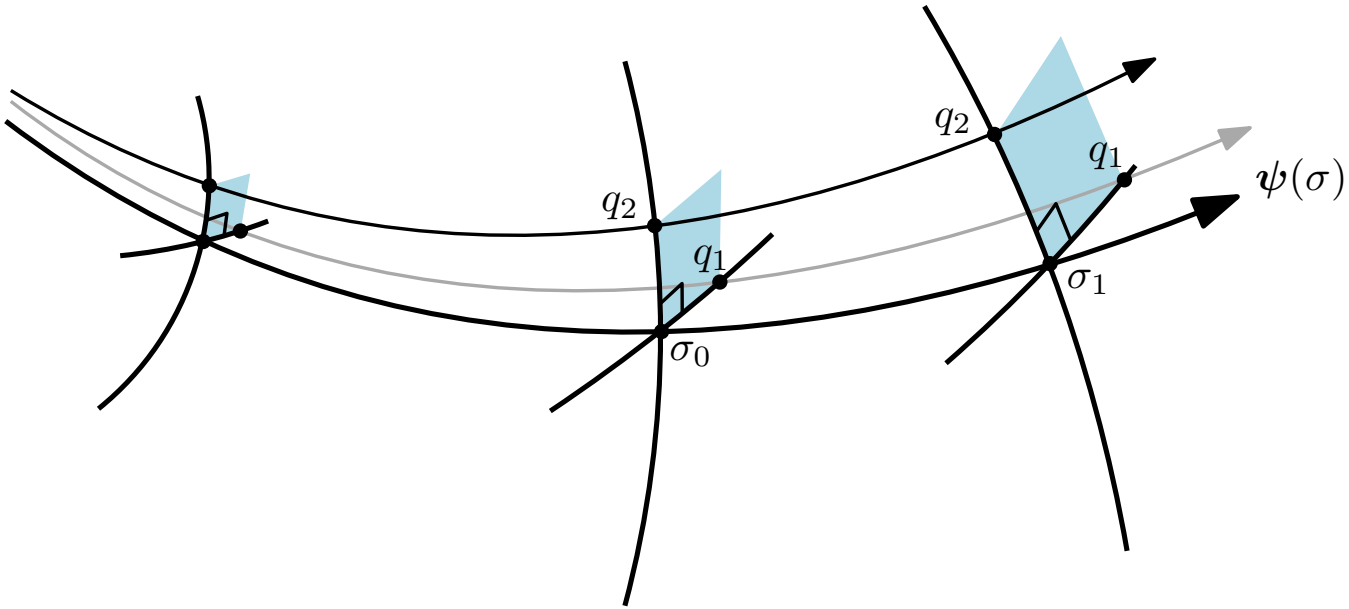
$$\Big(T(\hat{\boldsymbol{x}}) - T(\boldsymbol{x}_0)\Big)^2 + \Big(T(\hat{\boldsymbol{x}}) - T(\boldsymbol{x}_1)\Big)^2 = h^2 s(\hat{\boldsymbol{x}})^2$$

All `trial` nodes ordered in a priority queue by current T value

Sethian, 1996.

Introduction: the amplitude equation

Using **paraxial raytracing**, the amplitude can be computed in terms of the local geometric spreading of a ray tube:



The amplitude along a ray ψ satisfies:

$$\alpha(\psi(\sigma_1)) = \alpha(\psi(\sigma_0))\sqrt{\frac{s(\psi(\sigma_1))}{|\det(Q(\sigma_1))|}}, \qquad Q = \begin{bmatrix} q_1 & q_2 \end{bmatrix}$$

Popov, 1978.

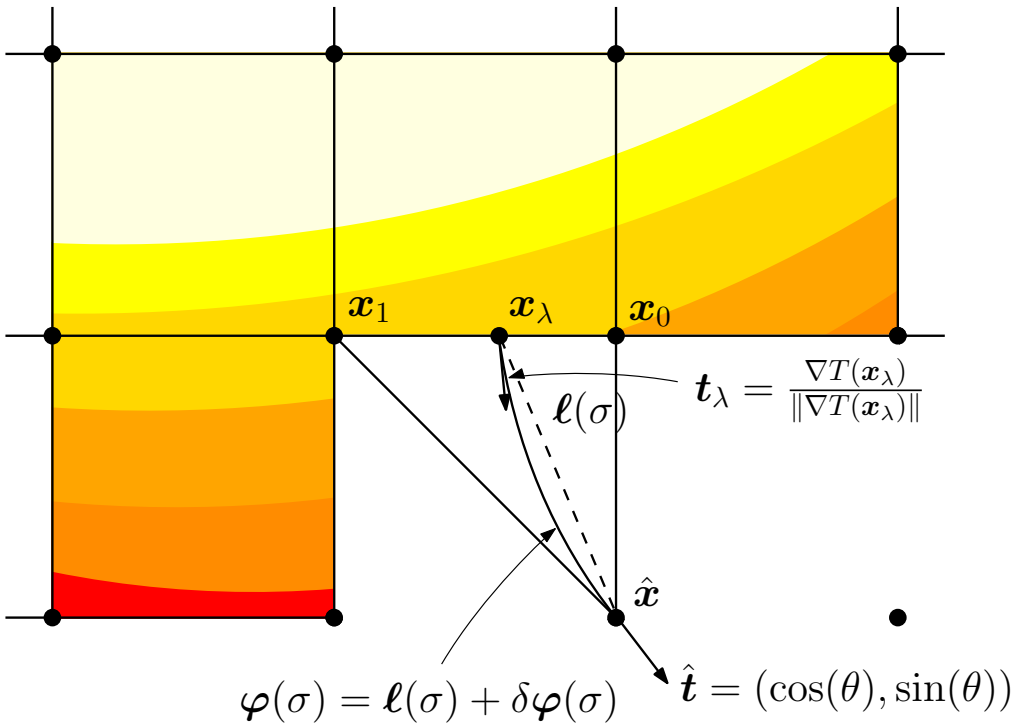
Jet marching methods (JMMs) in 2D

2D JMMs: overview

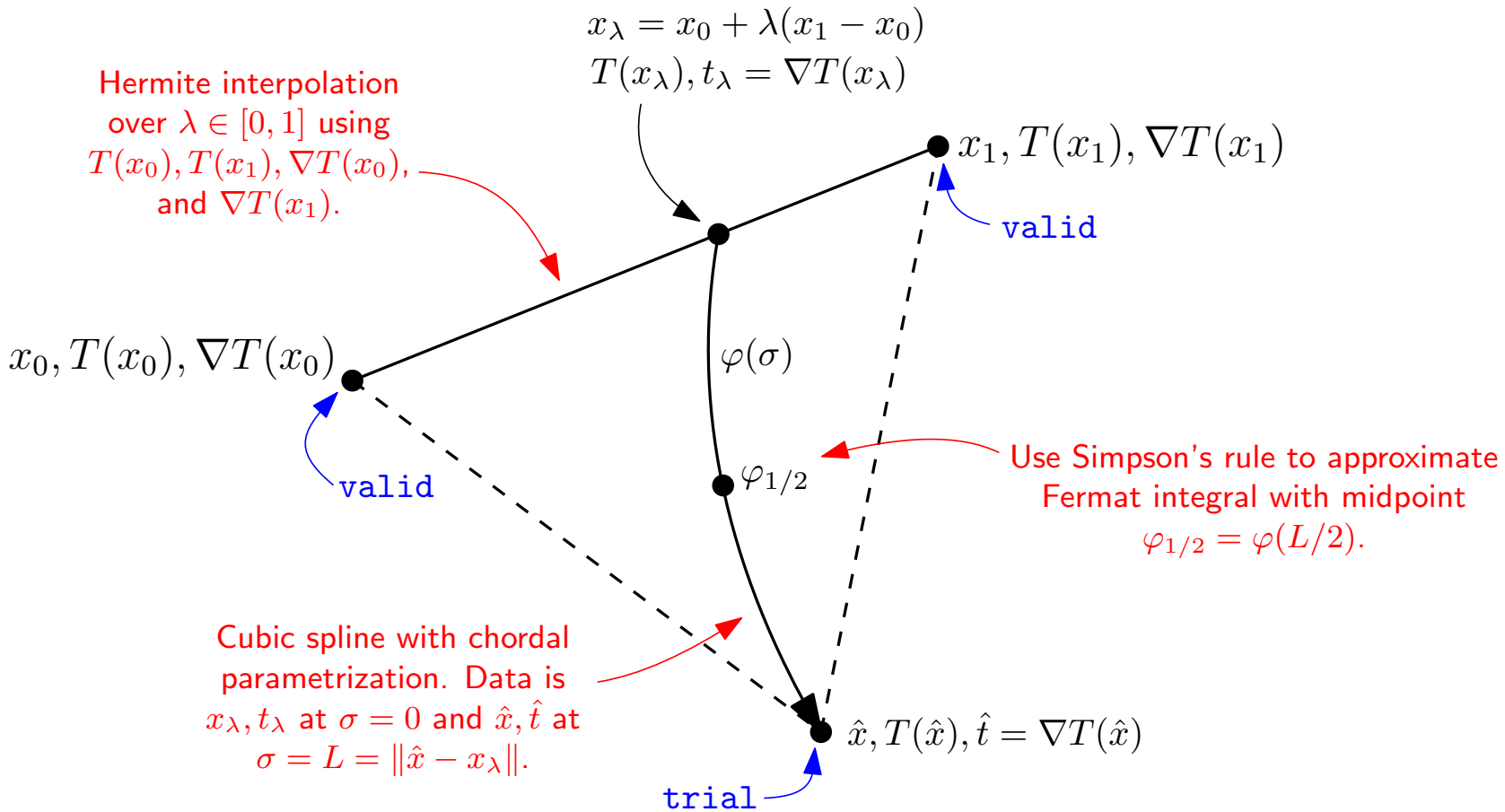
Approximately minimize Fermat's principle:

$$\tau(\hat{x}) = \min_{\psi} \left\{ \tau(\psi(0)) + \int_0^1 s(\psi(\sigma)) \|\psi'(\sigma)\| d\sigma \right\}$$

March the **jet** of the eikonal (coefficients of truncated Taylor polynomial):



2D JMMs: higher-order updates in 2D



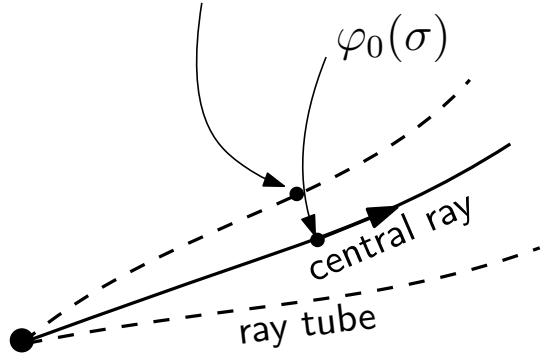
$$T(\hat{x}) = \min_{\substack{\hat{t}\in\mathbb{S}^1 \\ 0\le\lambda\le 1}} \left\{ \mathsf{T}(\lambda) + \frac{L}{6}\left(s(\hat{x}) + 4s(\varphi(\tfrac{L}{2}))\|\varphi'(\tfrac{L}{2})\| + s(x_\lambda) \right) \right\}$$

Potter and Cameron, 2021.

2D JMMs: marching the geometric spreading

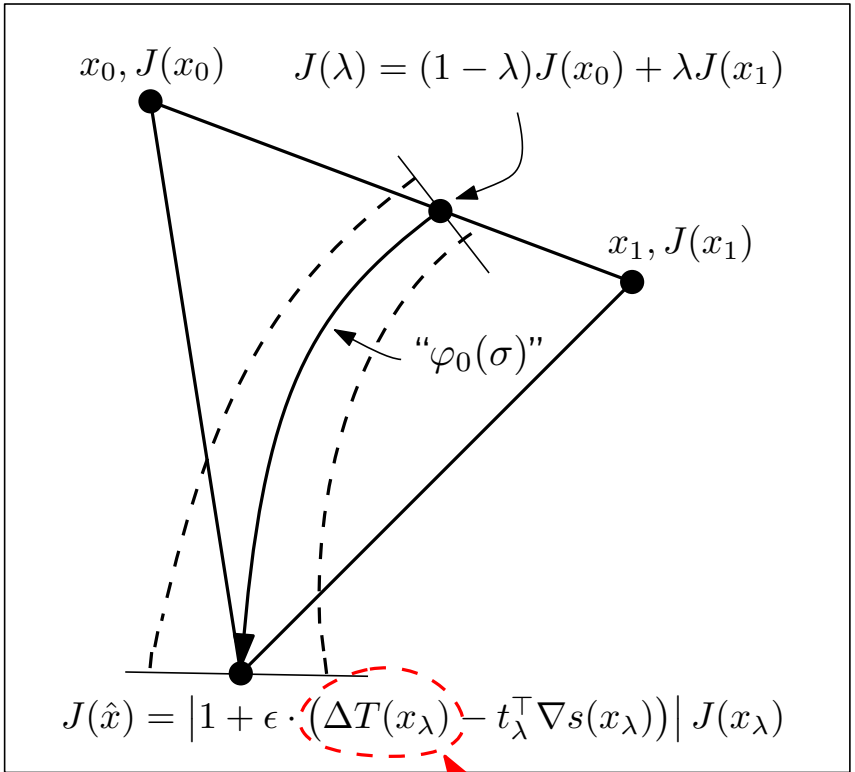
Linearly approximate the geometric spreading J over $[\boldsymbol{x}_0, \boldsymbol{x}_1]$ and march it after parametrizing the local ray:

$$\varphi(\sigma, q) = \varphi_0(\sigma) + E(\sigma)q$$
$$E^\top \varphi_0' \equiv 0$$



$$\begin{bmatrix} \dot{Q} \\ \dot{P} \end{bmatrix} = \begin{bmatrix} & c_0 I \\ \frac{-1}{c_0} \frac{\partial^2 c}{\partial q \partial q^\top}\Big|_{q=0} & \end{bmatrix} \begin{bmatrix} Q \\ P \end{bmatrix}$$

$$J = |\det Q|, \quad A(\phi_0(\sigma)) = \sqrt{\frac{c(\phi_0(\sigma))}{J(\phi_0(\sigma))}} A(\phi_0(0))$$

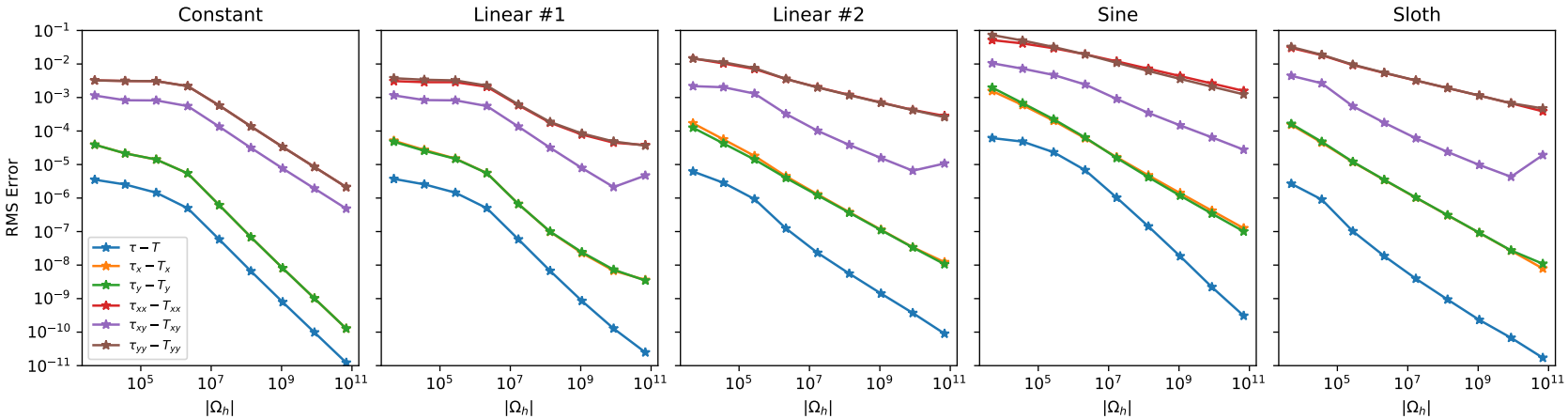


Need to know second partials of T at $x_\lambda \ldots$

Potter and Cameron, 2021.

2D JMMs: numerical results

Results using cell marching:



Order of convergence ($O(h^p)$) for jets:

	$\tau - T$	$\tau_x - T_x$	$\tau_y - T_y$	$\tau_{xx} - T_{xx}$	$\tau_{xy} - T_{xy}$	$\tau_{yy} - T_{yy}$
Constant	3.09	3.11	3.11	2.01	2.05	2.01
Linear #1	2.99	2.43	2.40	1.39	2.01	1.39
Linear #2	2.10	1.76	1.72	0.77	1.25	0.77
Sine	2.91	1.80	1.89	0.73	1.31	0.80
Sloth	2.03	1.76	1.75	0.75	1.33	0.76

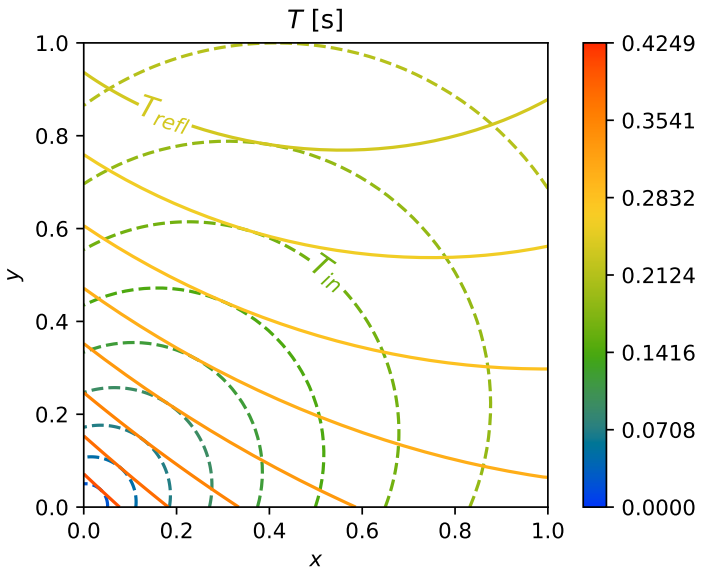
Potter and Cameron, 2021.

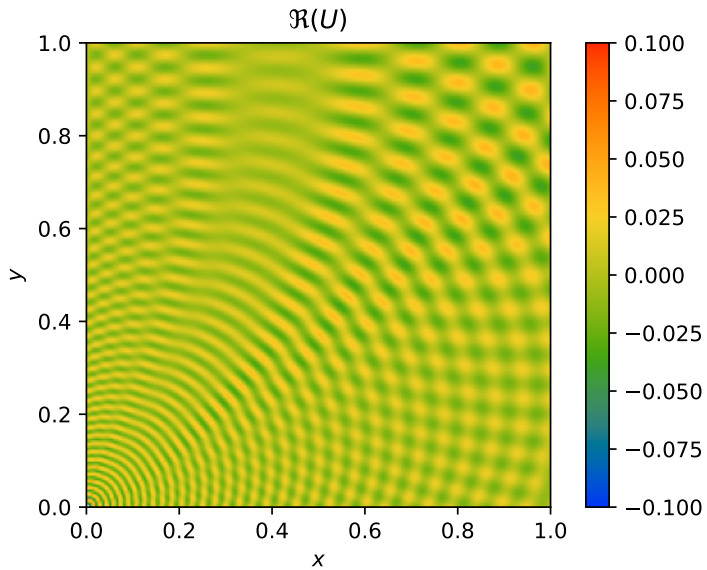
2D JMMs: reflections

Boundary conditions for reflected problem:

$$\tau_{\text{in}} = \tau_{\text{out}}, \quad J_{\text{in}} = J_{\text{out}} \quad \text{on wall}$$

and reflect $\nabla\tau_{\text{in}}$ over wall to get $\nabla\tau_{\text{out}}$:





2D JMMs: preliminary theoretical results

Theorem (JMM3 consistency)

Let $\Omega \subseteq \mathbb{R}^2$, assume that $s \in C^2(\Omega)$, and let $(\hat{\boldsymbol{x}}, \boldsymbol{x}_1, \boldsymbol{x}_2)$ be an update triangle where $\hat{\boldsymbol{x}}$ is `trial`*, and $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are* `valid`*. Assume that no caustic is incident on the update triangle, and that the ray arriving at $\hat{\boldsymbol{x}}$ crosses the segment $[\boldsymbol{x}_1, \boldsymbol{x}_2]$. Then:*

$$|\tau(\hat{\boldsymbol{x}}) - T(\hat{\boldsymbol{x}})| = O(h^4), \qquad \|\nabla\tau(\hat{\boldsymbol{x}}) - \nabla T(\hat{\boldsymbol{x}})\| = O(h^2). \tag{1}$$

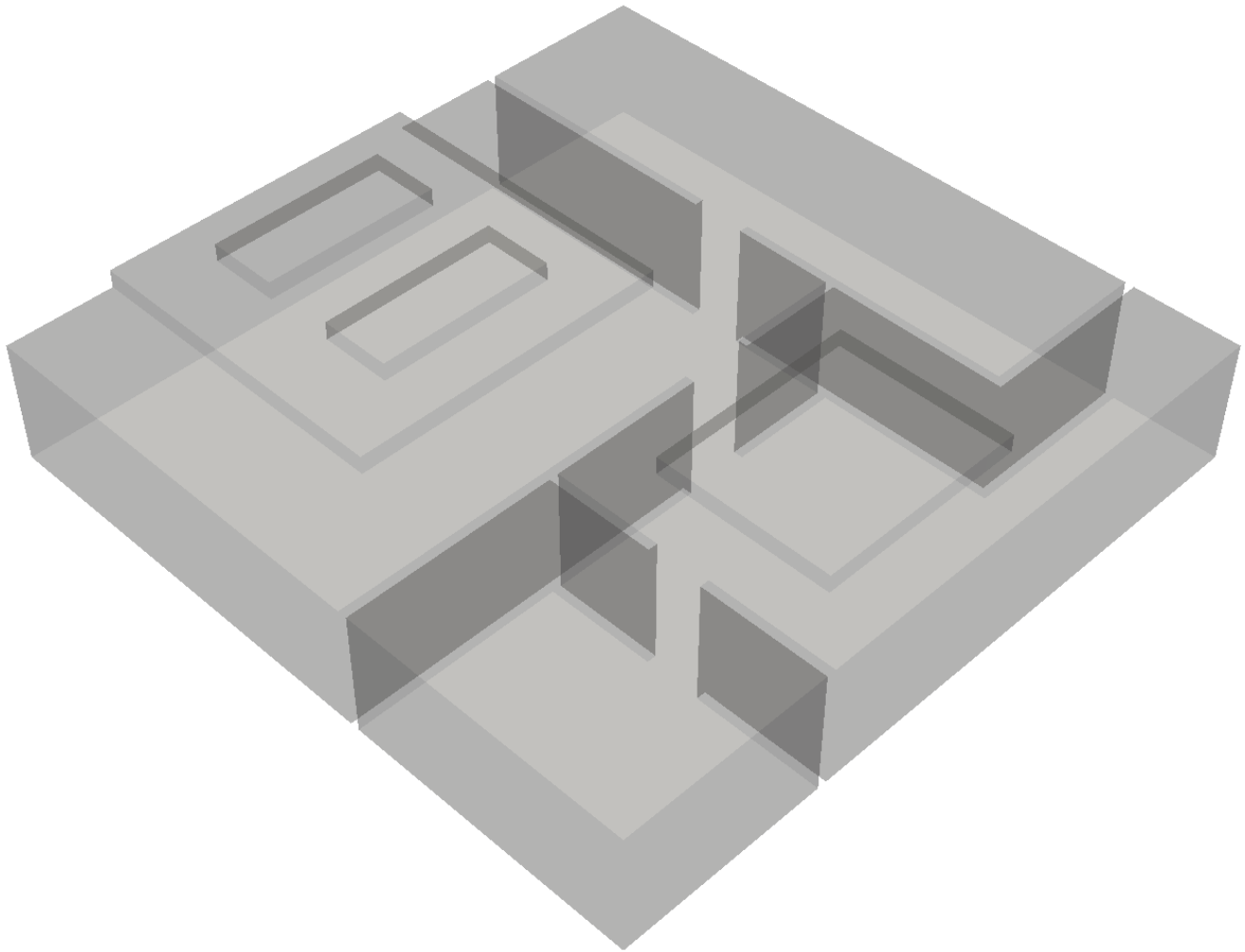
Theorem (full accuracy for $c \equiv 1$)

For a point source at the origin with $c \equiv 1$, if T and ∇T are initialized in a region surrounding the origin with $O(1)$ diameter, then the 2D JMM computes a solution such that $|\tau(\boldsymbol{x}) - T(\boldsymbol{x})| = O(h^3)$ and $\|\nabla\tau(\boldsymbol{x}) - \nabla T(\boldsymbol{x})\| = O(h^3)$ for each node $\boldsymbol{x}$ of a regular grid

A JMM for an unstructured tetrahedron mesh

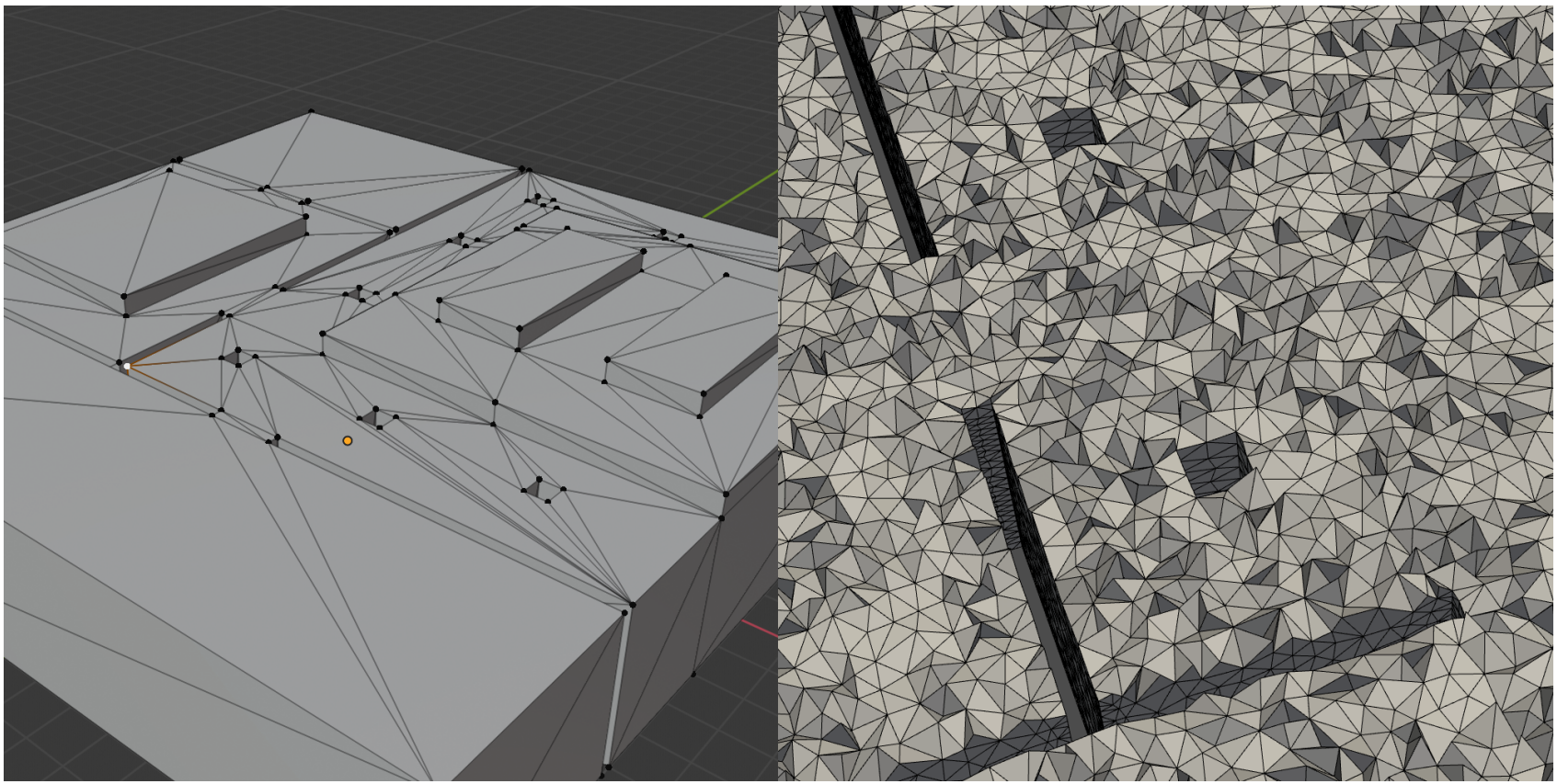
3D JMM: building test problem

Realistic test problem which includes some common architectural features, but not fully general (no small scale features, no curved surfaces, no floating obstacles, ...):



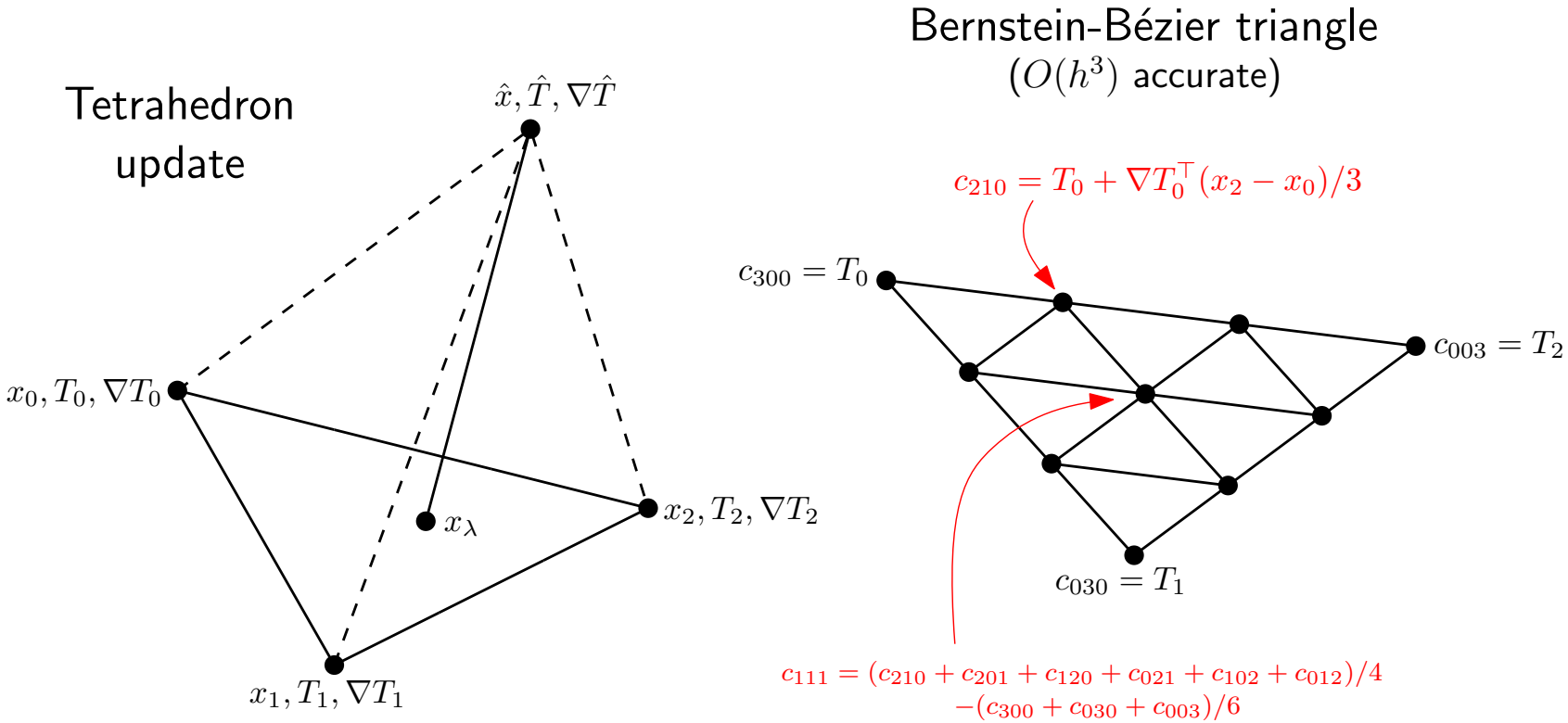
3D JMM: tetrahedron meshing

Modeled the test problem using Blender, discretized the boundary using tetgen:



3D JMM: tetrahedron updates

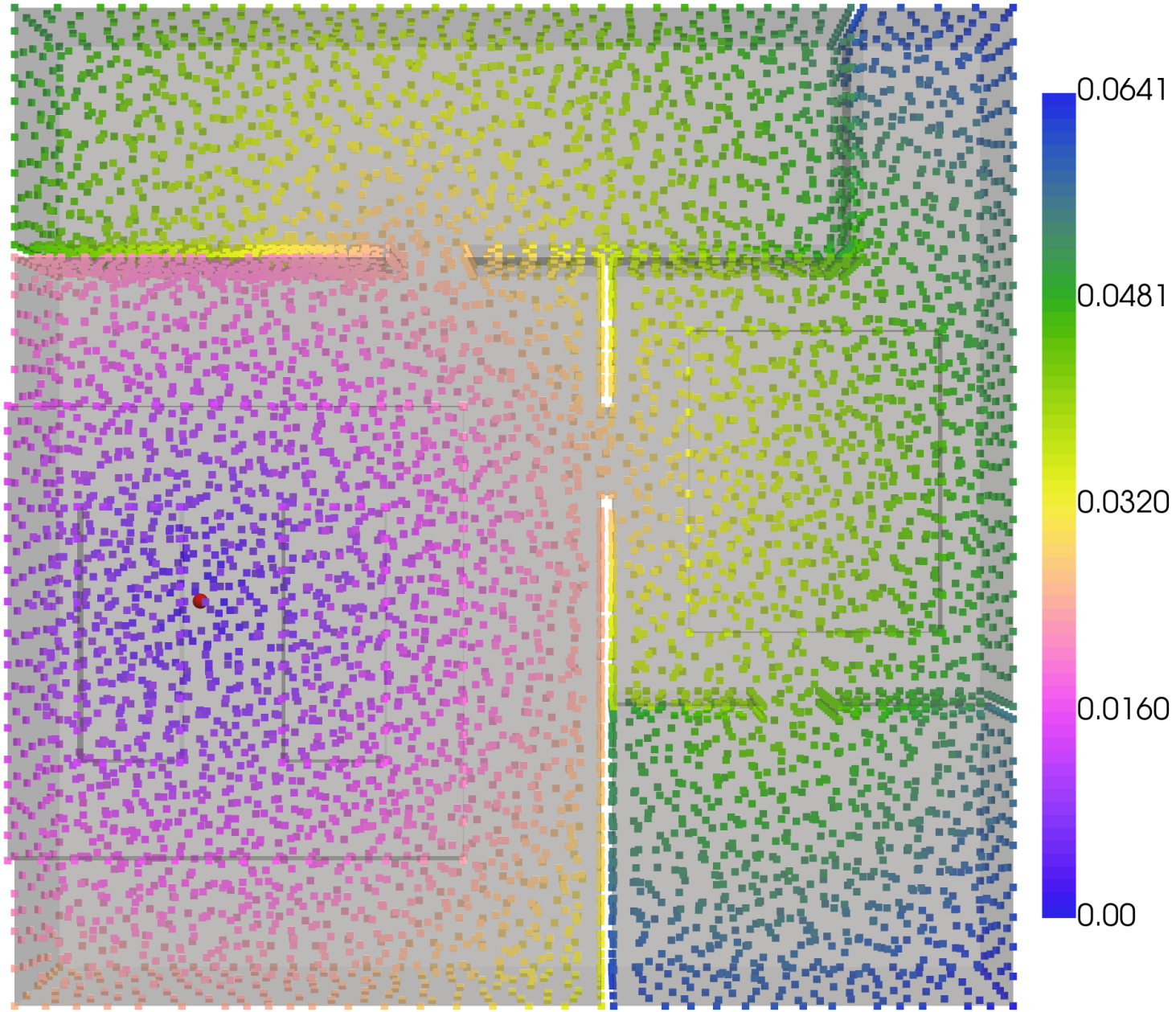
Workhorse for this solver is a cubically accurate tetrahedron update. Since $c \equiv 1$, rays are straight lines:



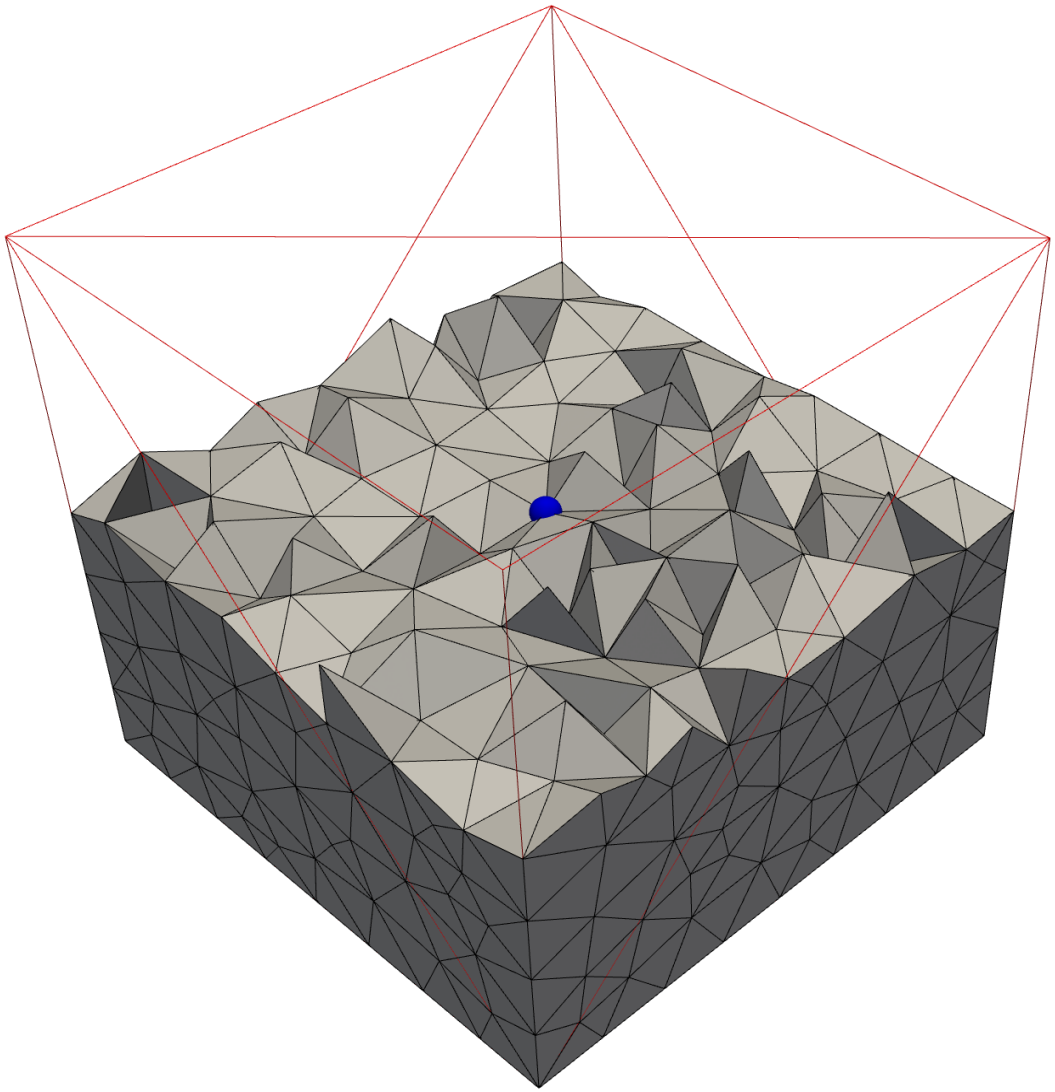
Minimize the cost function:

$$T(\hat{\boldsymbol{x}}) = \min_{\boldsymbol{\lambda}} \left\{ T(\boldsymbol{x}_{\boldsymbol{\lambda}}) + \|\hat{\boldsymbol{x}} + \boldsymbol{x}_{\boldsymbol{\lambda}}\| \right\}$$

3D JMM: eikonal on the test building domain



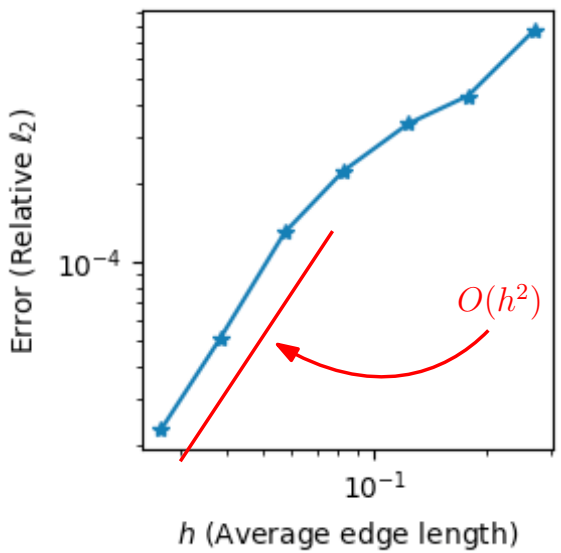
3D JMM: testing on a box domain



▲Input mesh

▲Tetrahedron mesh

▲Point source



3D JMM: movie

play movie

Multiple arrivals in 3D

MA in 3D: BCs for reflected and diffracted eikonal

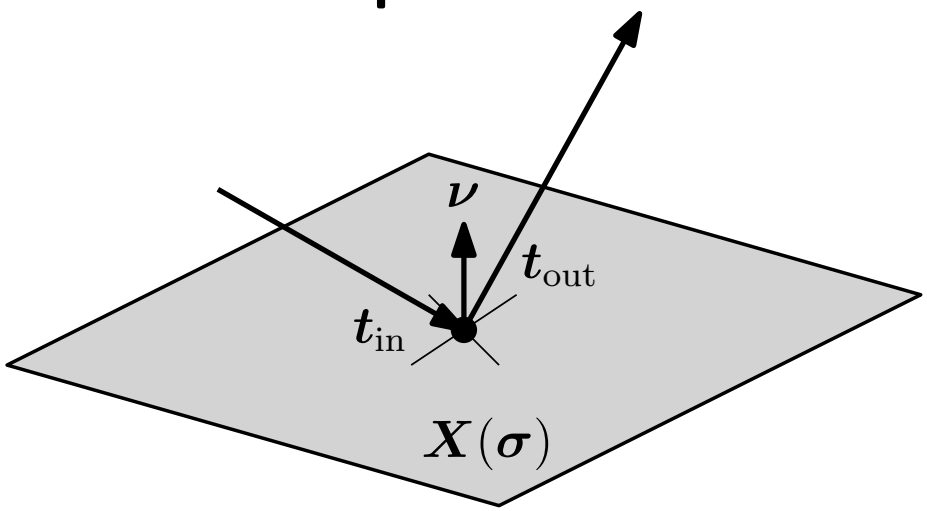
Boundary conditions for the reflected and diffracted eikonal:

$$\tau_{\text{in}}(\boldsymbol{X}(\boldsymbol{\sigma})) = \tau_{\text{out}}(\boldsymbol{X}(\boldsymbol{\sigma}))$$

Taking the gradient gives:

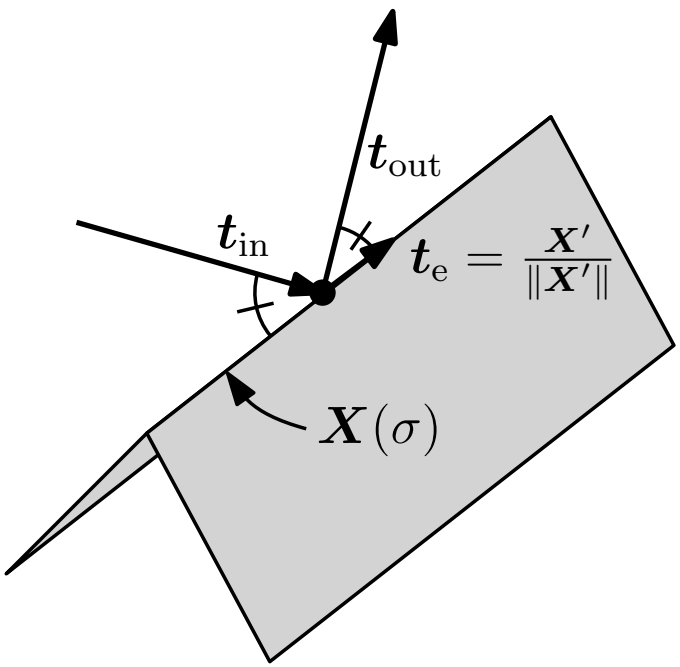
$$\nabla \boldsymbol{X}(\boldsymbol{\sigma})^\top \nabla \tau_{\text{in}}(\boldsymbol{X}(\boldsymbol{\sigma})) = \nabla \boldsymbol{X}(\boldsymbol{\sigma})^\top \nabla \tau_{\text{out}}(\boldsymbol{X}(\boldsymbol{\sigma}))$$

This leads to **specular reflection** and the **Keller cone**:



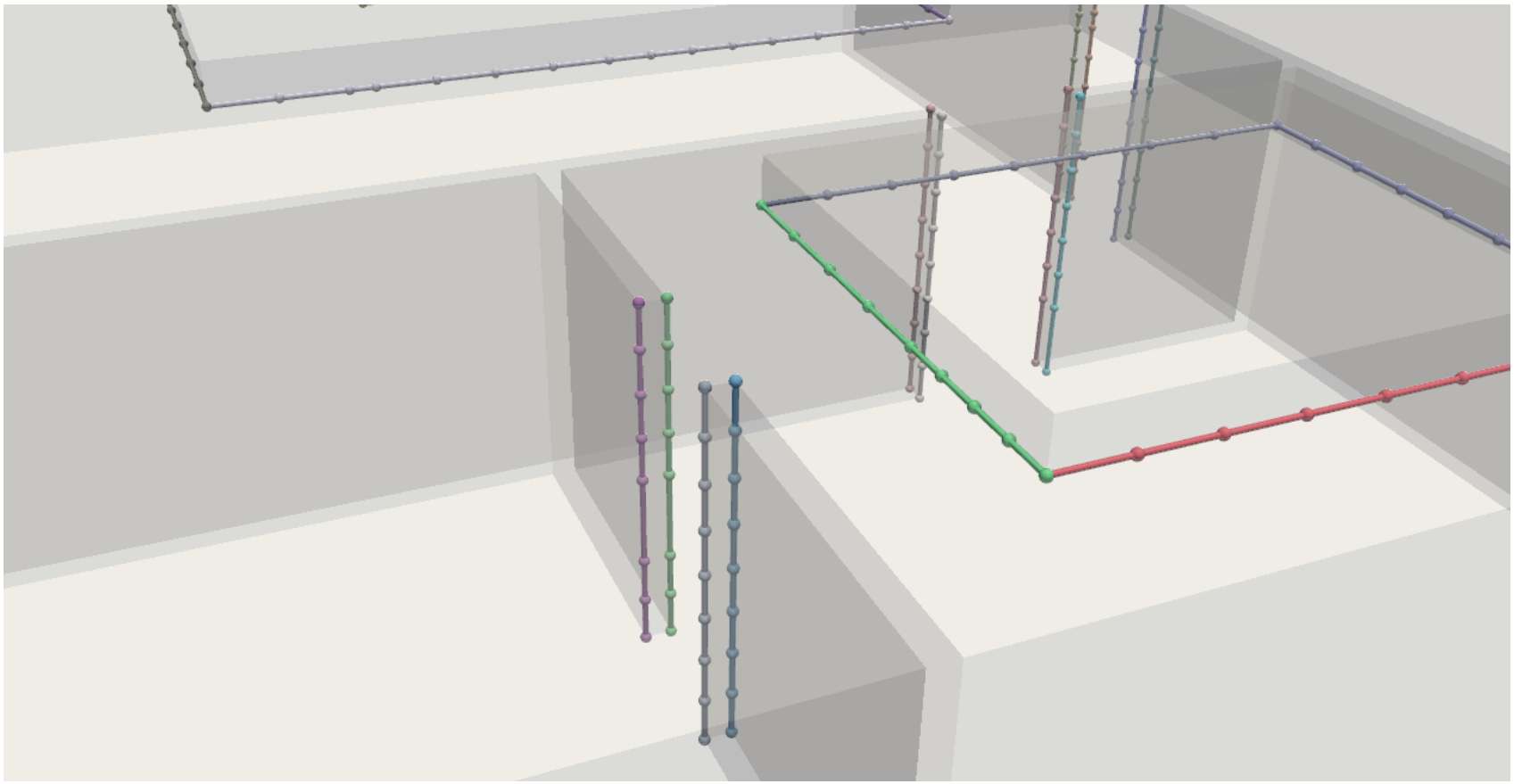
$$\boldsymbol{\nu} \in \text{null}(\boldsymbol{X})$$

$$\boldsymbol{t}_{\text{out}} = \left(\boldsymbol{I} - 2\boldsymbol{\nu}\boldsymbol{\nu}^\top\right)\boldsymbol{t}_{\text{in}}$$



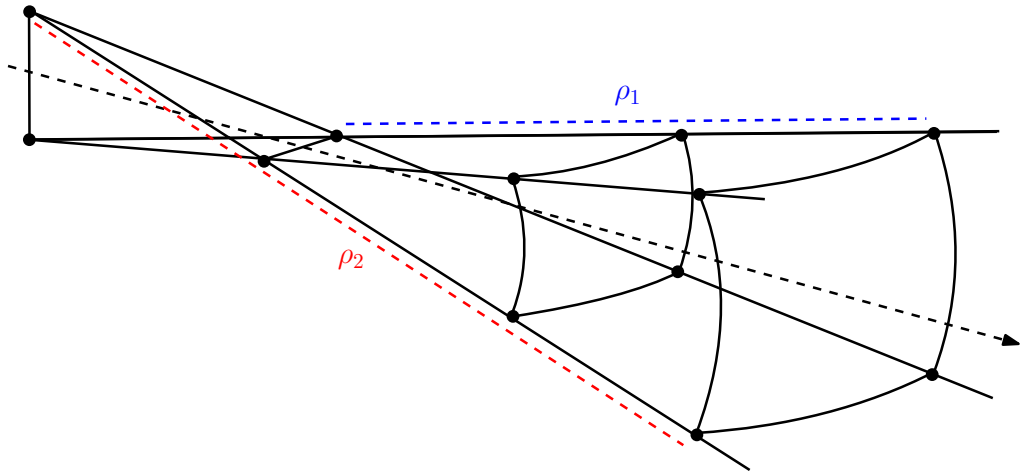
MA in 3D: reflectors and diffractors

Restrict τ_{in} to reflecting and diffracting subsets of $\partial\Omega$ and compute scattered fields:



MA in 3D: reflected and diffracted amplitudes

Astigmatic pencil of rays:



The amplitude for a reflection is:

$$\alpha_{\text{out}}(\boldsymbol{x}) = R(\boldsymbol{x}_0)\sqrt{\frac{\rho_1(\boldsymbol{x}_0)\rho_2(\boldsymbol{x}_0)}{\left[\rho_1(\boldsymbol{x}_0)+\sigma(\boldsymbol{x},\boldsymbol{x}_0)\right]\left[\rho_2(\boldsymbol{x}_0)+\sigma(\boldsymbol{x};\boldsymbol{x}_0)\right]}}\,\alpha_{\text{in}}(\boldsymbol{x}_0),$$

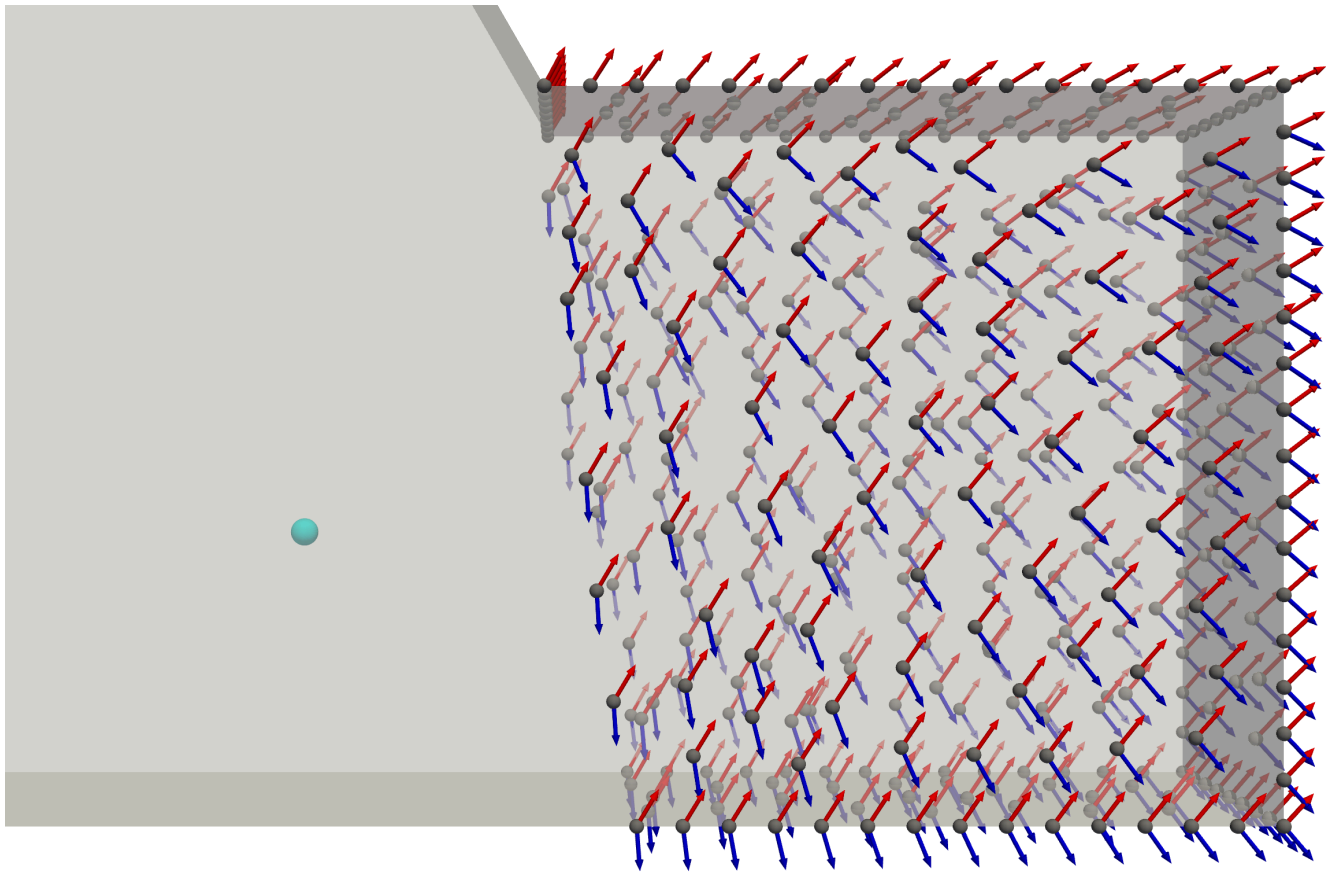
Amplitude for edge-diffracted ray is:

$$\alpha_{\text{out}}(\boldsymbol{x}) = D(\boldsymbol{x}_0)\sqrt{\frac{\rho_e(\boldsymbol{x}_0)}{\sigma(\boldsymbol{x};\boldsymbol{x}_0)\left[\rho_e(\boldsymbol{x}_0)+\sigma(\boldsymbol{x};\boldsymbol{x}_0)\right]}}\,\alpha_{\text{in}}(\boldsymbol{x}_0).$$

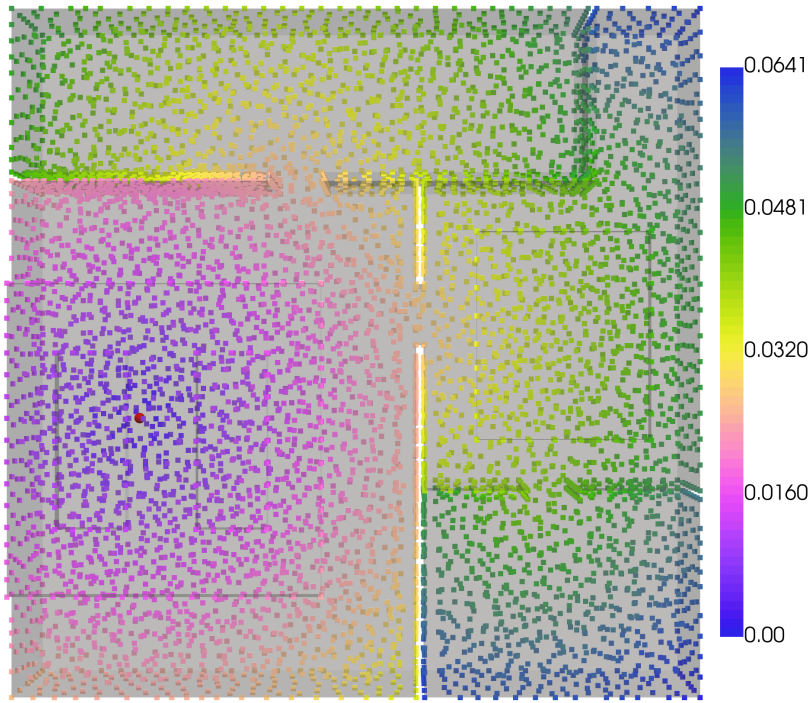
Complexity hidden in the UTD coefficient $D(\boldsymbol{x}_0)$.

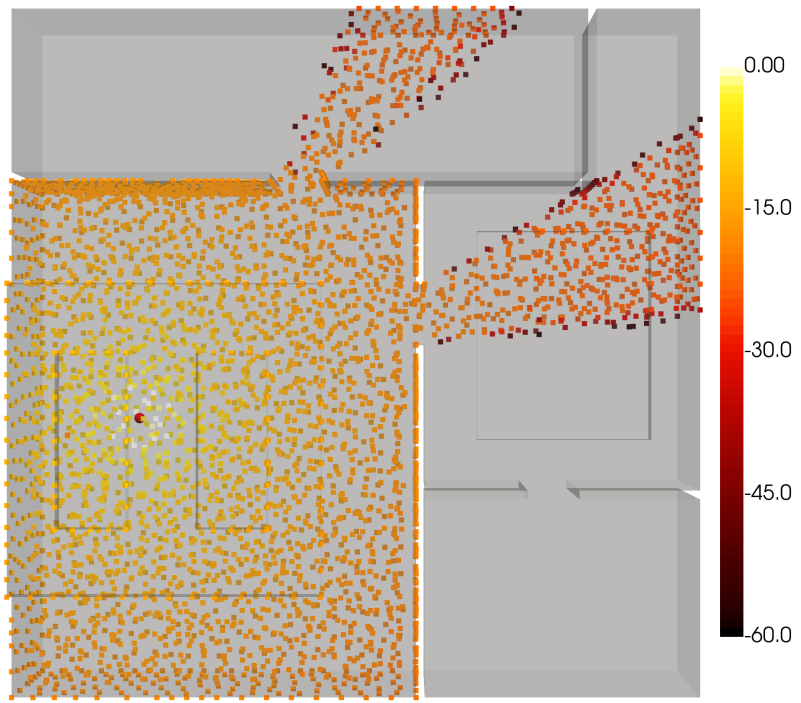
MA in 3D: big picture

Fast marching algorithms are dynamic programming algorithms. We can march auxiliary quantities along with the eikonal (such as the amplitude) in order to enforce BCs and determine the extent of GO fields.

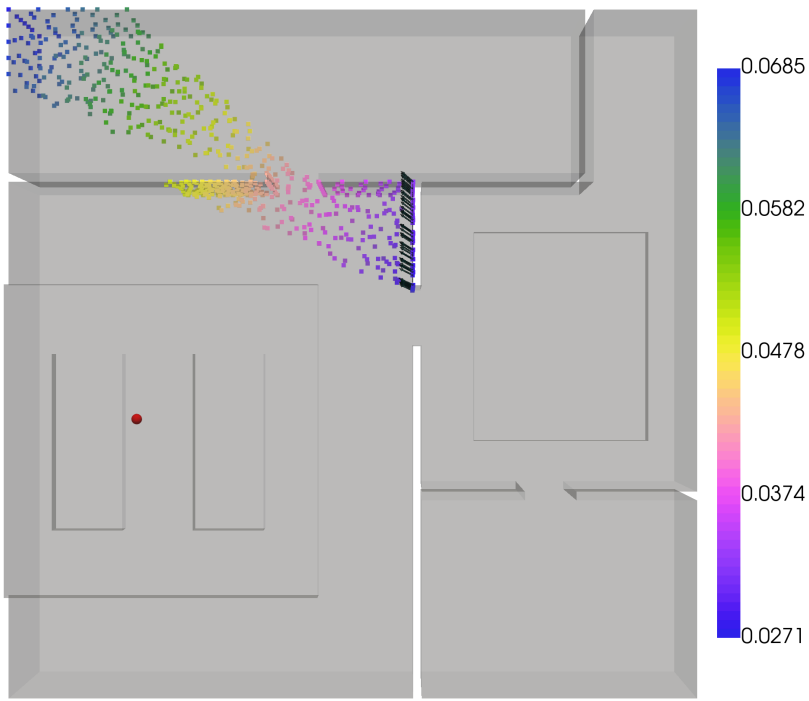


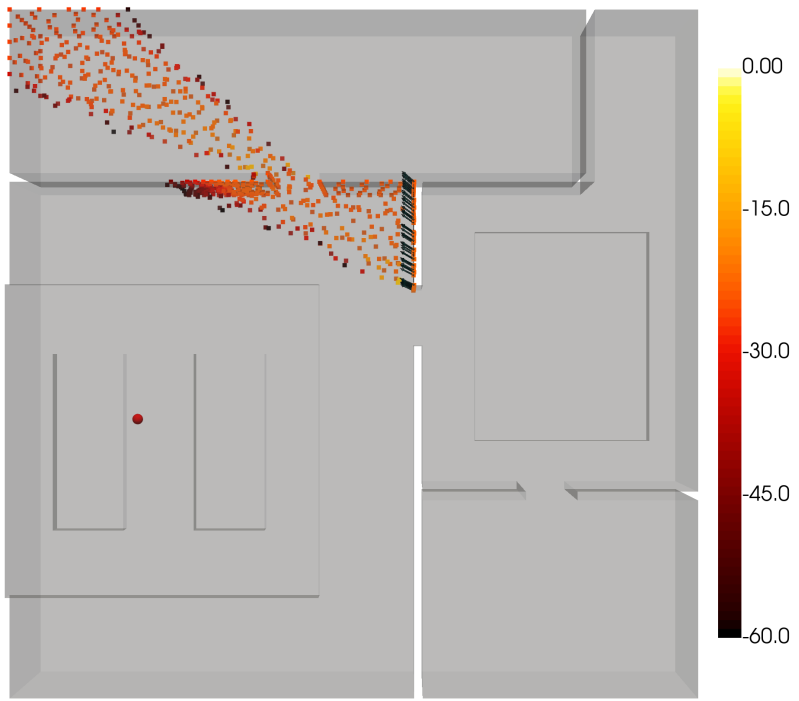
MA in 3D: building test problem



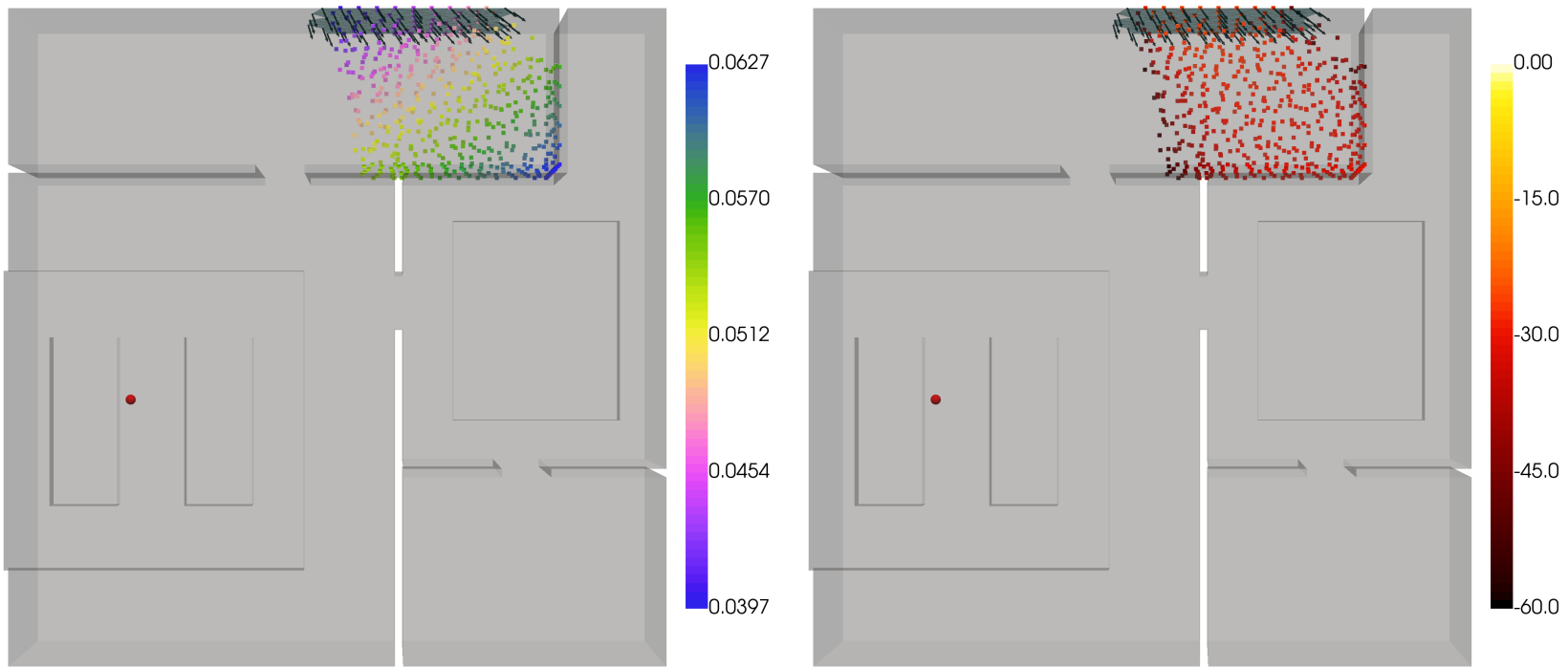


MA in 3D: building test problem

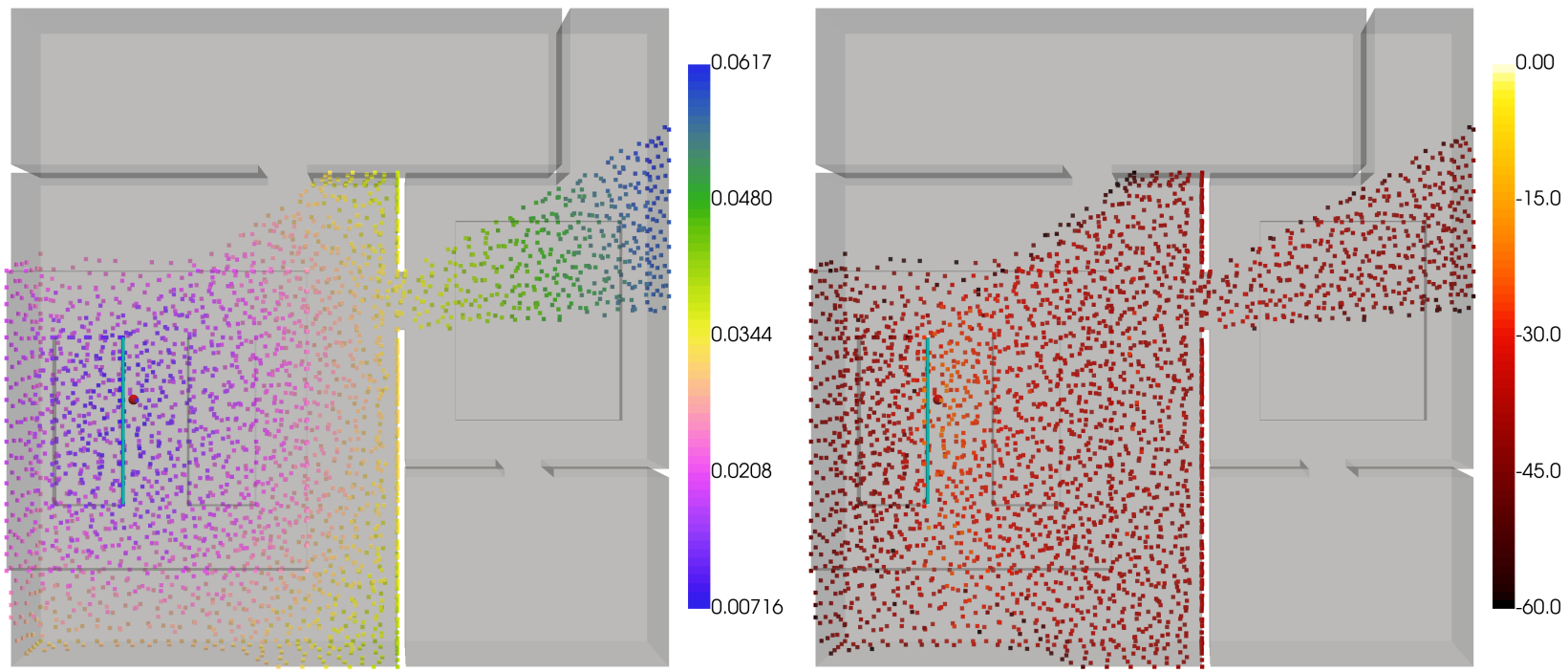




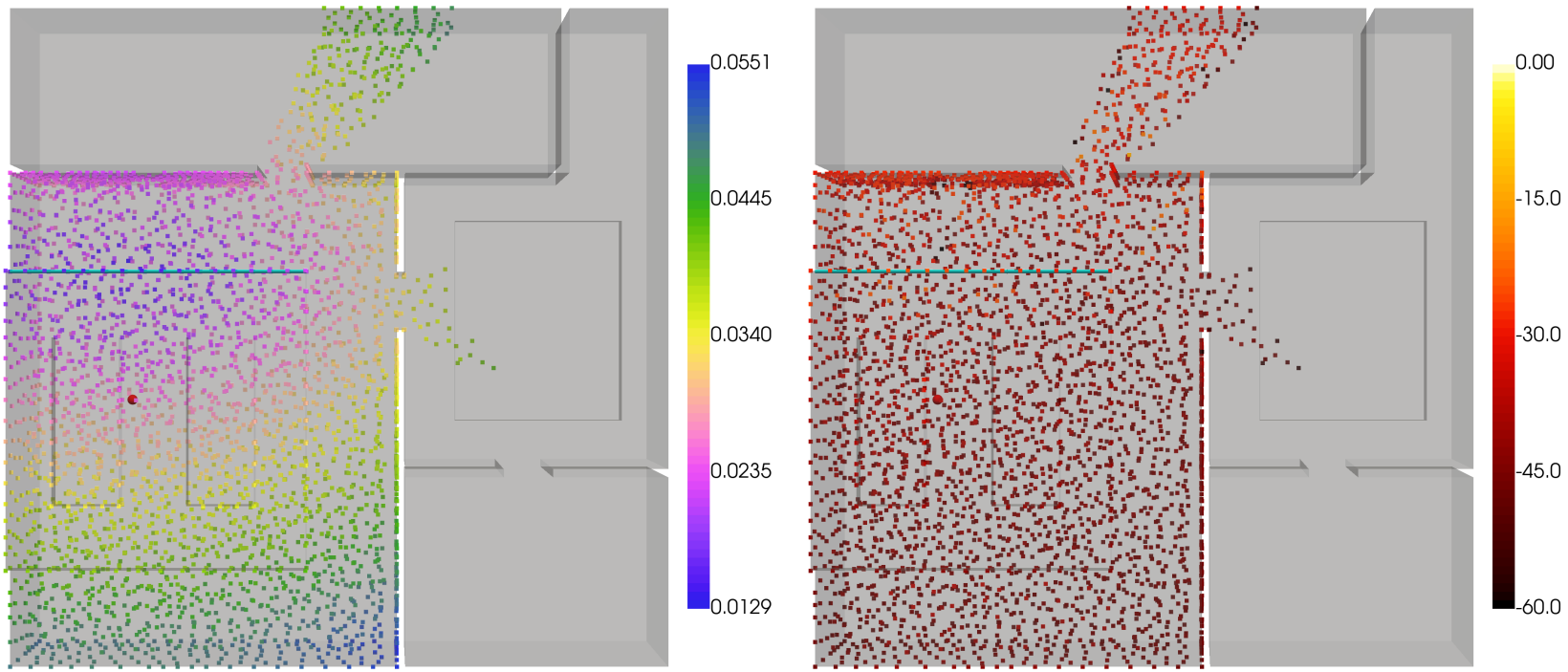
MA in 3D: building test problem



MA in 3D: building test problem



MA in 3D: building test problem

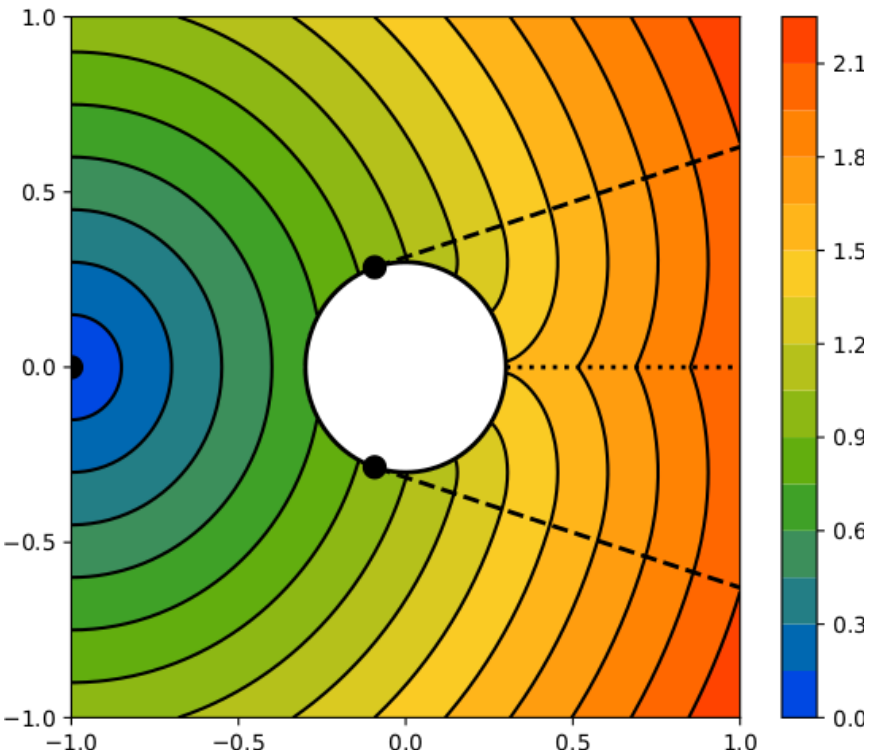


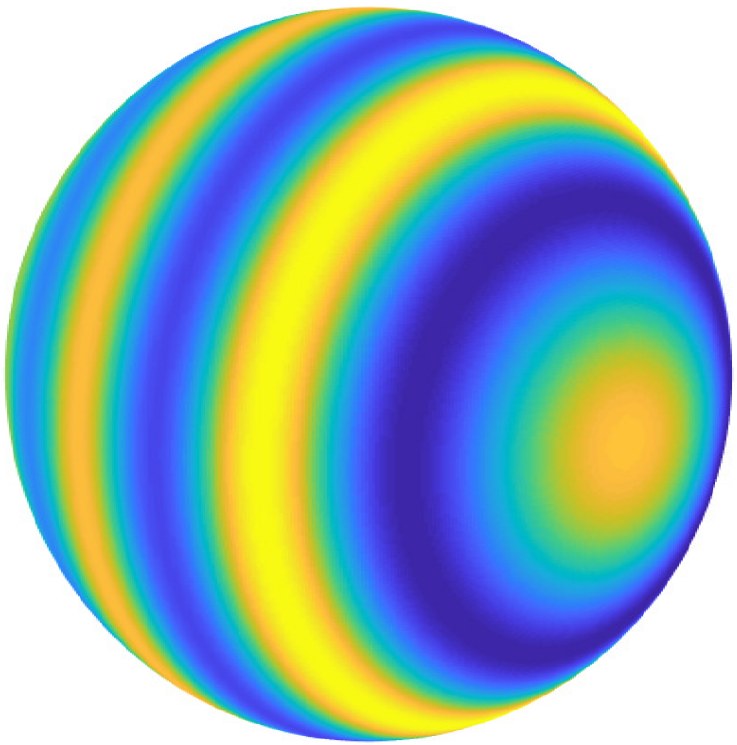
Conclusion and future work

Future work: scattering from a sphere

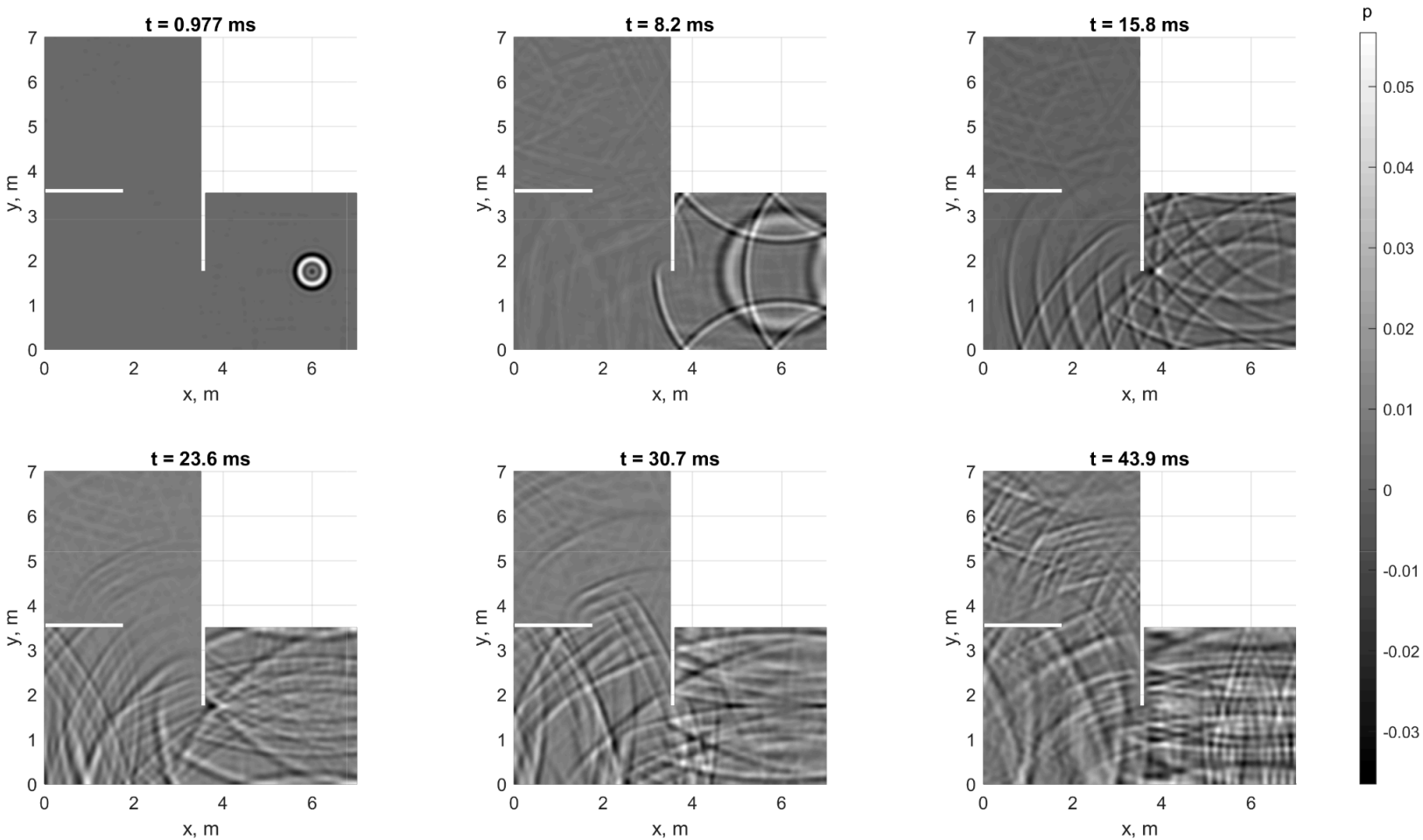
Analytic formula for sound hard scattering from a sphere:

$$P(\boldsymbol{x}) = ik \sum_{n=0}^{\infty} \sum_{m=-n}^{n} \left[j_n(kr) - \frac{j_n'(kr)}{h_n'(kr)} h_n(kr) \right] S_n^{-m}(\boldsymbol{x}_{\text{src}} - \boldsymbol{x}_0)\, Y_n^m(\boldsymbol{s})$$





Future work: comparison with other methods



Gumerov and Duraiswami, 2021.

Publications

Samuel F Potter, Maria K Cameron. *Ordered line integrals methods for solving the iekonal equation*, JOMP, 2020.

Samuel F Potter, Maria K Cameron. *Jet marching methods for solving the eikonal equation*, SISC, 2021.

Samuel F Potter, Maria K Cameron, Ramani Duraiswami. *Jet marching methods for high-frequency acoustic wave propagation*, in preparation.